

Question 1

```
from PIL import Image

width = 600

height = 400

image = Image.new("RGB", (width, height), "white")

image.show()

image.save("frame_buffer.png")
```

Question 2

```
from PIL import Image

image = Image.new("RGB", (400, 300), "white")

image.putpixel((50, 50), (255, 0, 0))    # Red

image.putpixel((150, 100), (0, 0, 255))  # Blue

image.putpixel((250, 200), (0, 255, 0))  # Green

image.putpixel((350, 250), (0, 0, 0))    # Black

image.show()
```

Question 3

```
from PIL import Image

image = Image.new("RGB", (400, 300), "white")

def draw_pixel(image, x, y, colour):

    for i in range(x, x + 20):

        for j in range(y, y + 20):

            image.putpixel((i, j), colour)

draw_pixel(image, 50, 50, (255, 0, 0))    # Red

draw_pixel(image, 150, 100, (0, 0, 255))  # Blue

draw_pixel(image, 250, 150, (0, 255, 0))  # Green

image.show()
```

Question 4

```
from PIL import Image
```

```
image = Image.new("RGB", (400, 300), "white")

for x in range(50, 351):
    image.putpixel((x, 100), (255, 0, 0))

for y in range(50, 251):
    image.putpixel((200, y), (0, 0, 255))

for x in range(50, 251):
    y = x
    image.putpixel((x, y), (0, 255, 0))

image.show()
```

Question 5

```
from PIL import Image

image = Image.new("RGB", (500, 300), "white")

for x in range(50, 201):
    image.putpixel((x, 50), (255, 0, 0))

for x in range(50, 201):
    image.putpixel((x, 150), (255, 0, 0))

for y in range(50, 151):
    image.putpixel((50, y), (255, 0, 0))

for y in range(50, 151):
    image.putpixel((200, y), (255, 0, 0))

for x in range(250, 401):
    image.putpixel((x, 100), (0, 0, 255))

for x in range(250, 401):
    image.putpixel((x, 250), (0, 0, 255))

for y in range(100, 251):
    image.putpixel((250, y), (0, 0, 255))

for y in range(100, 251):
    image.putpixel((400, y), (0, 0, 255))
```

```
image.show()
```

Question 6

```
from PIL import Image
```

```
image = Image.new("RGB", (400, 300), "white")
```

```
orange = (255, 165, 0)
```

```
for x in range(100, 301):
```

```
    for y in range(100, 251):
```

```
        image.putpixel((x, y), orange)
```

```
image.show()
```

Question 7

```
from PIL import Image
```

```
image = Image.new("RGB", (600, 300), "white")
```

```
for x in range(200):
```

```
    red = int((x / 199) * 255)
```

```
    for y in range(300):
```

```
        image.putpixel((x, y), (red, 0, 0))
```

```
for x in range(200, 400):
```

```
    position = x - 200
```

```
    green = int((position / 199) * 255)
```

```
    for y in range(300):
```

```
        image.putpixel((x, y), (0, green, 0))
```

```
for x in range(400, 600):
```

```
    position = x - 400
```

```
    blue = int((position / 199) * 255)
```

```
    for y in range(300):
```

```
    image.putpixel((x, y), (0, 0, blue))  
image.show()
```

Question 8 — AMFEX

```
from PIL import Image  
  
def rectangle(x1, y1, x2, y2, colour):  
    for x in range(x1, x2 + 1):  
        for y in range(y1, y2 + 1):  
            image.putpixel((x, y), colour)  
  
red = (255, 0, 0)  
blue = (0, 0, 255)  
green = (0, 255, 0)  
orange = (255, 165, 0)  
purple = (150, 0, 200)  
  
# LETTER A  
rectangle(50, 50, 110, 70, red)  
rectangle(50, 70, 70, 180, red)  
rectangle(90, 70, 110, 180, red)  
rectangle(70, 100, 90, 120, red)  
  
# LETTER M  
rectangle(180, 50, 200, 180, blue)  
rectangle(200, 50, 220, 80, blue)  
rectangle(220, 70, 240, 100, blue)  
rectangle(240, 50, 260, 80, blue)  
rectangle(260, 50, 280, 180, blue)  
  
# LETTER F  
rectangle(350, 50, 370, 180, green)  
rectangle(370, 50, 430, 70, green)  
rectangle(370, 100, 420, 120, green)
```

LETTER E

rectangle(480, 50, 500, 180, orange)

rectangle(500, 50, 560, 70, orange)

rectangle(500, 100, 550, 120, orange)

rectangle(500, 160, 560, 180, orange)

LETTER X

rectangle(620, 50, 640, 80, purple)

rectangle(680, 50, 700, 80, purple)

rectangle(640, 80, 660, 120, purple)

rectangle(660, 80, 680, 120, purple)

rectangle(640, 120, 680, 150, purple)

rectangle(620, 150, 640, 180, purple)

rectangle(680, 150, 700, 180, purple)

image.show()

image.save("AMFEX_pixel_art.png")